

Vinh Ly

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Summary

Software Engineer with **18 years** of experience building large-scale distributed systems and data infrastructure. Owned end-to-end systems serving billions of requests daily at Google and YouTube, including video processing pipelines and recommendation services. Led teams of 5-12 engineers across multiple time zones while managing on-call rotations for critical production services. Built real-time data platforms processing millions of events per second and developed internal tools that reduced developer friction. Seeking to apply deep backend and infrastructure expertise to AI-powered product development at Zillow. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages and Frameworks: TypeScript, JavaScript, HTML, CSS, SQL

Distributed Systems: Kubernetes, Kafka, Spanner, Bigtable, MapReduce, Flume

Data Infrastructure: Pub/Sub, BigQuery, Dataflow, Cloud Spanner, Druid, Storm

Backend Development: C++, Java, Python, Go, MapReduce, gRPC

Machine Learning Platforms: TensorFlow, ML Training Infrastructure, Feature Stores, A/B Testing

Work Experience

Staff Software Engineer | YouTube | 02/2025 - Present

- Led migration of legacy data pipelines to modern stream processing architecture, enabling real-time analytics for creator monetization features with p99 latency under 100ms.
- Architected video processing pipeline improvements handling millions of uploads daily, reducing transcoding latency by 25% through optimized resource allocation and parallelization strategies.
- Partnered with product managers on A/B testing infrastructure, reducing experiment iteration cycle from 2 weeks to 3 days for new feature rollouts.
- Designed multi-region deployment strategy for recommendation services serving 2B+ users, improving availability during regional outages from 99.9% to 99.99%.
- Mentored 3 junior engineers on distributed systems design and conducted technical interviews, improving team hiring pipeline throughput by 40%.

Staff Software Engineer | Whatnot | 05/2024 - 02/2025

- Established on-call rotation and incident response protocols, decreasing mean time to resolution from 45 minutes to 12 minutes for P1 incidents.
- Built real-time bidding system processing 100K+ concurrent auctions, implementing event-driven architecture that reduced system latency by 35% during peak traffic.
- Designed inventory management microservices handling 500K+ daily transactions, achieving 99.99% uptime through comprehensive circuit breakers and graceful degradation.
- Led security audit of payment processing pipeline, identifying and remediating 12 critical vulnerabilities before SOC 2 compliance review.
- Reduced cloud infrastructure costs by 22% through resource right-sizing and spot instance adoption for batch workloads.

Staff Software Engineer/Manager | Google | 03/2015 - 04/2024

- Built automated model deployment pipeline handling 500+ daily releases, eliminating manual deployment errors and reducing rollout time from 4 hours to 15 minutes.
- Designed TensorFlow distributed training framework supporting 10K+ GPU clusters, enabling model training time reduction from days to hours for production models.
- Managed team of 8 engineers building core ML training infrastructure powering search ranking and advertising systems serving 4B+ users globally.

- Owned Bigtable-based feature store serving 1M+ queries per second with p99 latency under 10ms, critical for real-time ranking systems.
- Led cross-functional initiative to unify ML experimentation platforms across 15 product teams, reducing duplicate infrastructure maintenance overhead by 60%.

Software Engineer | Google | 02/2011 - 02/2015

- Built real-time monitoring system using Storm and Druid processing 100M+ events per hour, providing sub-second visibility into ads system performance.
- Optimized BigQuery query performance for ad hoc analytics, reducing query costs by 30% through partitioning and clustering strategies.
- Mentored 2 incoming engineers on data infrastructure and codebase, accelerating their time-to-productivity from 6 months to 2 months.
- Developed MapReduce data processing pipelines analyzing 50PB+ daily log data for search quality metrics and ranking experiments.
- Designed A/B testing framework supporting 1000+ concurrent experiments, enabling rapid iteration on search ranking improvements.

Software Design Engineer II | Microsoft | 10/2007 - 12/2011

- Created automated testing framework reducing QA cycle time by 40% through parallel test execution and continuous integration.
- Developed backend services for enterprise collaboration platform processing 10M+ daily API requests with 99.99% availability requirements.
- Designed database schema migrations for SQL Azure supporting 100GB+ datasets with zero downtime deployment strategy.
- Built notification delivery system handling 5M+ push notifications daily using Windows Azure cloud infrastructure.
- Implemented single sign-on authentication flows serving 500K+ enterprise users across multiple identity providers.

Software Design Engineer Intern | Microsoft | 06/2006 - 08/2006

- Developed internal tooling for test case management system used by 200+ developers across product teams.
- Built automated performance benchmarking suite identifying 15 critical bottlenecks in data access layer.

Education

B.S. Honors in Computer Science

Virginia Tech | 2004 - 2007